In Chapter 3 we also discussed the fate of light emitted from the surfaces of static stars. We learned that because time flows more slowly at the stellar surface than far away *(gravitational time dilation)*, light waves emitted from the star's surface and received far away will have a lengthened period of oscillation and correspondingly a lengthened wavelength and a redder color. The light's wavelength gets shifted toward the red end of the spectrum as the light climbs out of the star's intense gravitational field *(gravitational redshift)*. When the static star is four times larger than its critical circumference, the light's wavelength is lengthened by 15 percent (see the photon of light in the upper right part of the figure); when the star is at twice its critical circumference, the redshift is 41 percent (middle right); and when the star is precisely at its critical circumference, the light's wavelength is infinitely redshifted, which means that the light has no energy left at all and therefore has ceased to exist.

Oppenheimer and Snyder, in their quick calculation, inferred two things from this sequence of static stars: First, an imploding star, like these static stars, would probably develop strong spacetime curvature as it nears its critical circumference, but not infinite curvature and therefore not infinite tidal gravitational forces. Second, as the star implodes, light from its surface should get more and more redshifted, and when it reaches the critical circumference, the redshift should become infinite, making the star become completely invisible. In Oppenheimer's words, the star should "cut itself off' visually from our external Universe.

Was there any way, Oppenheimer and Snyder asked themselves, that the star's internal properties—ignored in this quick calculation—could save the star from this cutoff fate? For example, might the implosion be forced to go so slowly that never, even after an infinite time, would the critical circumference actually be reached?

Oppenheimer and Snyder would have liked to answer these questions by calculating the details of a realistic stellar implosion, as depicted in the left half of Figure 6.3. Any real star will spin, as does the Earth, at least a little bit. Centrifugal forces due to that spin will force the star's equator to bulge out at least a little bit, as does Earth's equator. Thus, the star cannot be precisely spherical. As it implodes, the star must spin faster and faster like a figure skater pulling in his arms; and its faster spin will cause centrifugal forces inside the star to grow, making the equatorial bulge more pronounced—sufficiently pronounced, perhaps, that it even halts the